

Machine Learning-Gated NQ Futures Strategy: Walk-Forward Validation, Risk Controls, and Failure Analysis

Technical portfolio report - refreshed with conservative metrics

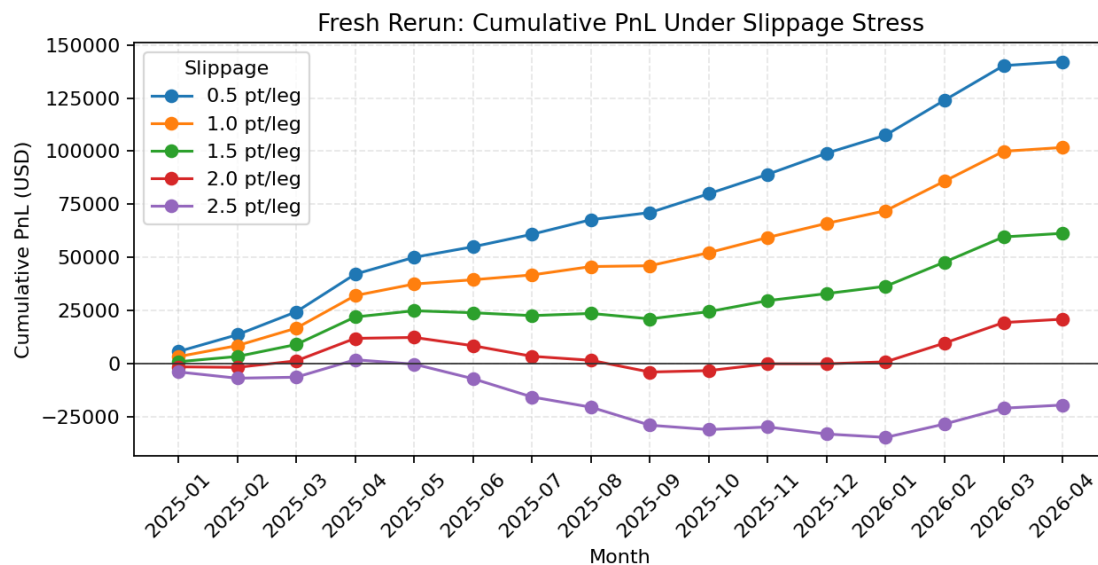
Problem. The first weakness in this project was the headline metric. The old 9+ number was a trade-level t-statistic, not a finance-style Sharpe ratio. This report uses the fresh June 5, 2026 rerun, monthly annualized Sharpe, slippage stress, walk-forward folds, and explicit failure cases.

Data. The fresh validation used Databento-derived NQ 1-minute bars from the original research workspace. The Phase 0 rerun `phase0\_20260605T225254Z` executed four scripts successfully, then `src/fresh\_conservative\_validation.py` saved timestamped trade streams for portfolio-grade metrics.

Portfolio headline. The believable claim is the stressed result: at 1.5 NQ points of slippage per leg, the filtered strategy produced 2,022 trades, +\$61,301 PnL, 66.6% win rate, profit factor 1.38, monthly annualized Sharpe 2.85, and 13/16 positive months. At 2.0 points it became marginal; at 2.5 points it lost money.

Overfitting audit. An independent rebuild reproduced the original conservative metrics and added cleaner validation. The clean-eval rerun produced +\$60,511 at 1.5 pt/leg, and the purged clean-eval rerun produced +\$56,145. That supports the NQ-specific research result, but it does not prove a live or broadly general futures edge.

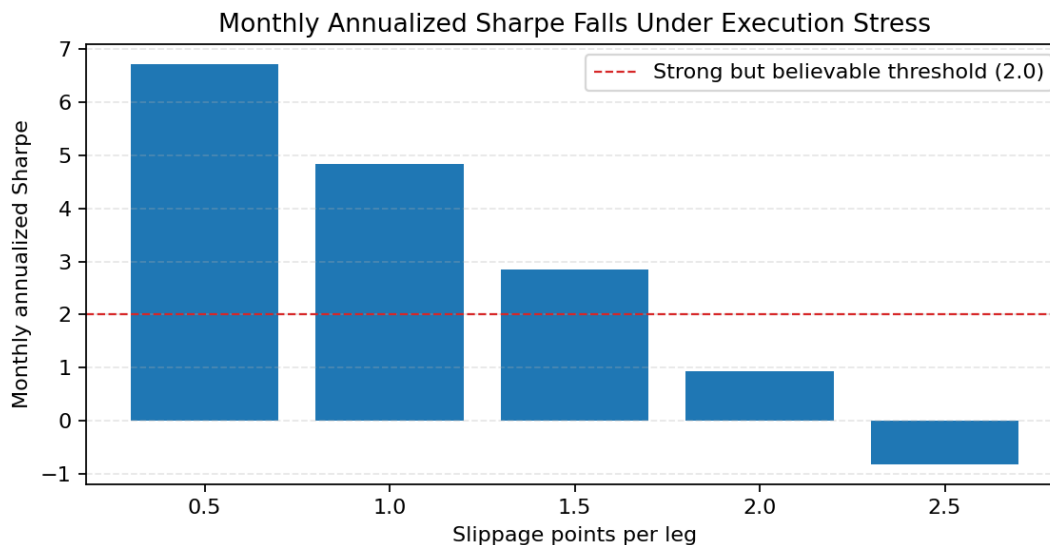
Variant	Trades	PnL	Win	Monthly Sharpe	Positive Months
Base modeled 0.5 pt/leg	2,022	+\$142,181	69.9%	6.71	16/16
Stress 1.5 pt/leg	2,022	+\$61,301	66.6%	2.85	13/16
Stress 2.0 pt/leg	2,022	+\$20,861	63.2%	0.93	9/16
Stress 2.5 pt/leg	2,022	-\$19,579	58.6%	-0.81	6/16
Directional comparison	2,022	-\$6,275	28.4%	-0.48	8/16



Method

The tested strategy is an ML-gated synthetic straddle. When the volatility gate fires, the backtest enters both long and short exposure around an NQ bar and exits with asymmetric ATR-based profit target and stop loss logic. The method does not claim reliable direction prediction; the directional comparison is included specifically to show that direction alone failed.

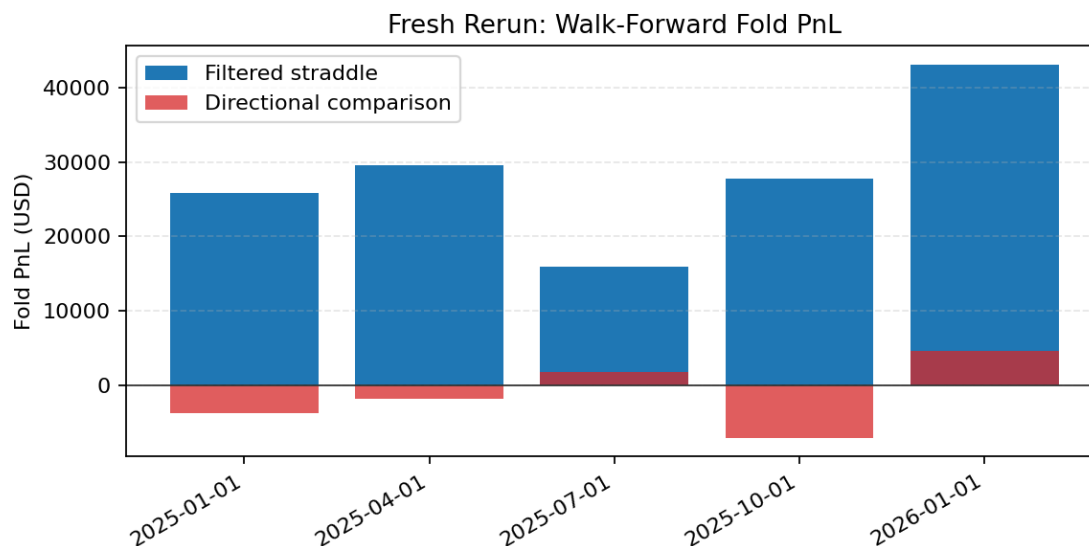
- Walk-forward design: 12-month train, 3-month test, 3-month slide.
- Fresh rerun command: ``python phase0_freeze_runner.py``, then ``python src/fresh_conservative_validation.py``.
- Independent audit command: ``python src/overfit_replication_audit.py``.
- Conservative metric change: old ``sh`` is reported only as ``trade_t_stat_not_sharpe``; headline metrics use monthly PnL.
- Execution stress: extra cost is subtracted per trade for 1.0, 1.5, 2.0, and 2.5 NQ points per leg.



The stress curve is the point of the project. The strategy looks excellent at the modeled base cost, remains portfolio-worthy at 1.5 points per leg, becomes weak at 2.0, and fails at 2.5. That boundary makes the result more believable because it admits execution quality can kill the edge.

Walk-Forward Results

The fresh rerun produced 5 positive filtered-straddle test folds. Fold PnL ranged from \$15,963 to \$43,101. This is useful evidence, but five folds is still not enough to claim a live-trading edge without forward paper trading.



Delayed-entry stress was not damaging in this rerun: the delayed-entry variant produced \$+145,548 with monthly Sharpe 6.57. That is encouraging, but it is still a bar-based simulation, not broker fill evidence.

The harshest execution finding came from same-bar adverse ordering: at 1.5 pt/leg it produced \$-175,264. That means intrabar fill ordering is a live-risk blocker unless broker-level replay or paper fills confirm the optimistic assumption.

Failure Analysis

The key failure case is directional prediction. On the same filtered trade count, the directional comparison lost \$6,275, had only 28.4% win rate, and monthly Sharpe -0.48. That supports the story that the work is about volatility/risk gating, not a clean NQ direction model.

- Execution-cost failure: the strategy loses money at 2.5 points of slippage per leg.
- Model-scope failure: the independent same-logic cross-instrument audit failed on ES and MGC.
- Metric-label failure: the old 9+ and 13+ values are trade-level t-statistics, not headline Sharpe ratios.
- Deployment failure risk: no broker fills or locked forward paper-trading period have been proven yet.

Instrument	Trades	PnL	Monthly Sharpe
ES	251	\$-11,847	-3.33
MGC	228	\$-9,625	-3.10

Limitations And Next Steps

This is a legitimate research pipeline, but the claims must stay bounded. The backtests assume historical bar data, modeled slippage/commission, clean fills within bar ranges, and no live order-routing errors. The ML gate may still overfit time-of-day and volatility structure. The strongest result is specific to NQ/equity-index behavior.

- Run a locked forward paper-trading period with no parameter changes.
- Compare simulated entries/exits against NinjaTrader or broker paper fills.
- Add hard daily loss limits, news-event exclusions, and volatility-regime shutdown rules.
- Expand validation to more futures contracts before claiming broad generalization.
- Keep the 1.5 pt/leg stress case as the resume/report headline; mention the base case only as supporting evidence.

Portfolio takeaway

The strongest truthful story is that this project built a futures research workflow: market data cleaning, feature engineering, ML-gated filtering, walk-forward testing, slippage stress, drawdown-aware parameter search, failure analysis, and reproducible reporting. It should not be described as a proven profitable trading bot.

Reproduction commands: `python src/reproduce_metrics.py`; fresh rerun via `python src/fresh_conservative_validation.py`